

■ Crypto ML Project — Scoring Report

CortisolTeam — Final Model Evaluation

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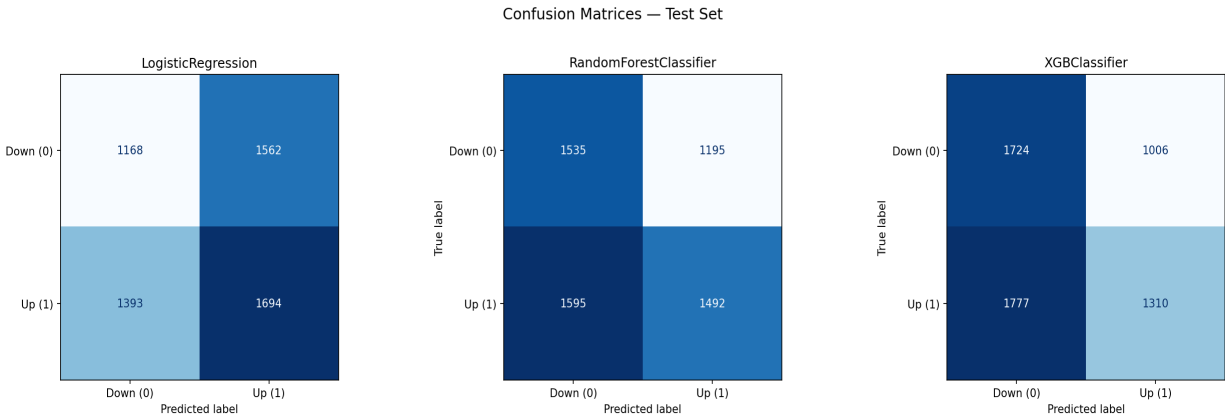
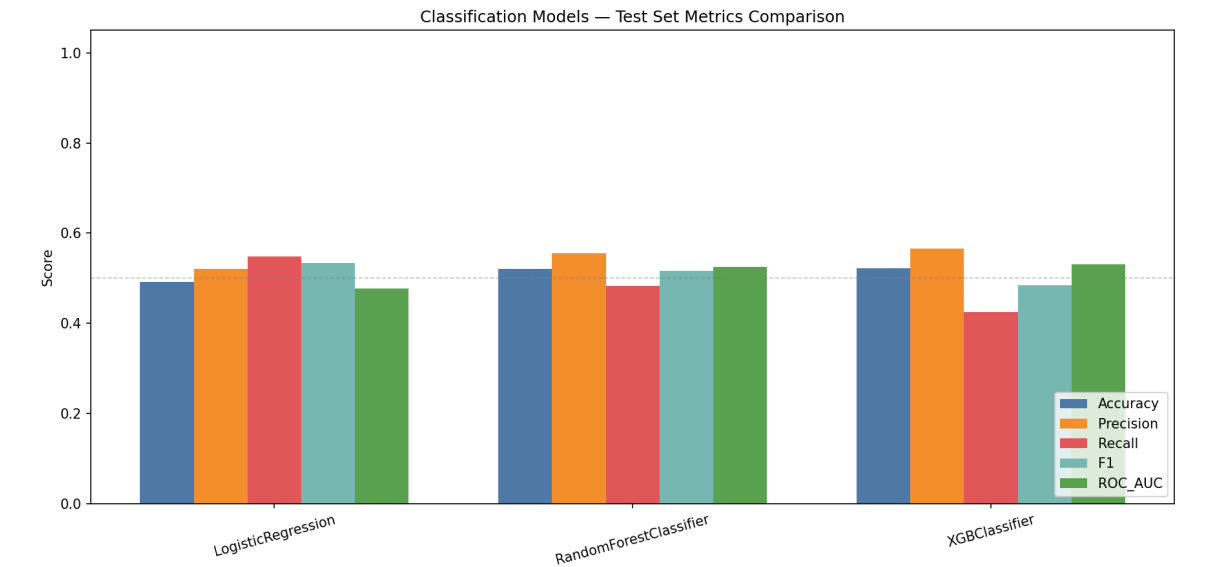
This report consolidates the evaluation of all supervised and unsupervised models trained on cryptocurrency historical price data. It includes classification, regression, and clustering metrics, confusion matrices, ROC curves, and a simple backtesting simulation.

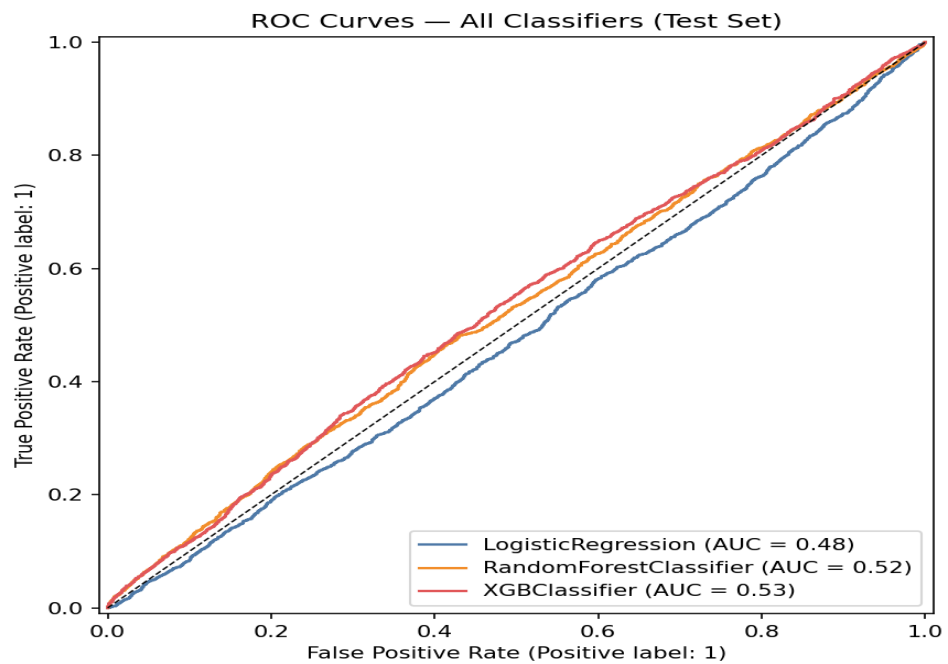
1. Classification Models

Three classifiers were trained to predict whether the next-day close price will increase (1) or decrease (0). Evaluation uses temporal train/val/test split (80/10/10) with TimeSeriesSplit cross-validation.

Classification — Test Set Metrics

	Accuracy	Precision	Recall	F1	ROC-AUC
LogisticRegression	0.492	0.5203	0.5488	0.5341	0.4769
RandomForestClassifier	0.5204	0.5553	0.4833	0.5168	0.5244
XGBClassifier	0.5216	0.5656	0.4244	0.4849	0.5305



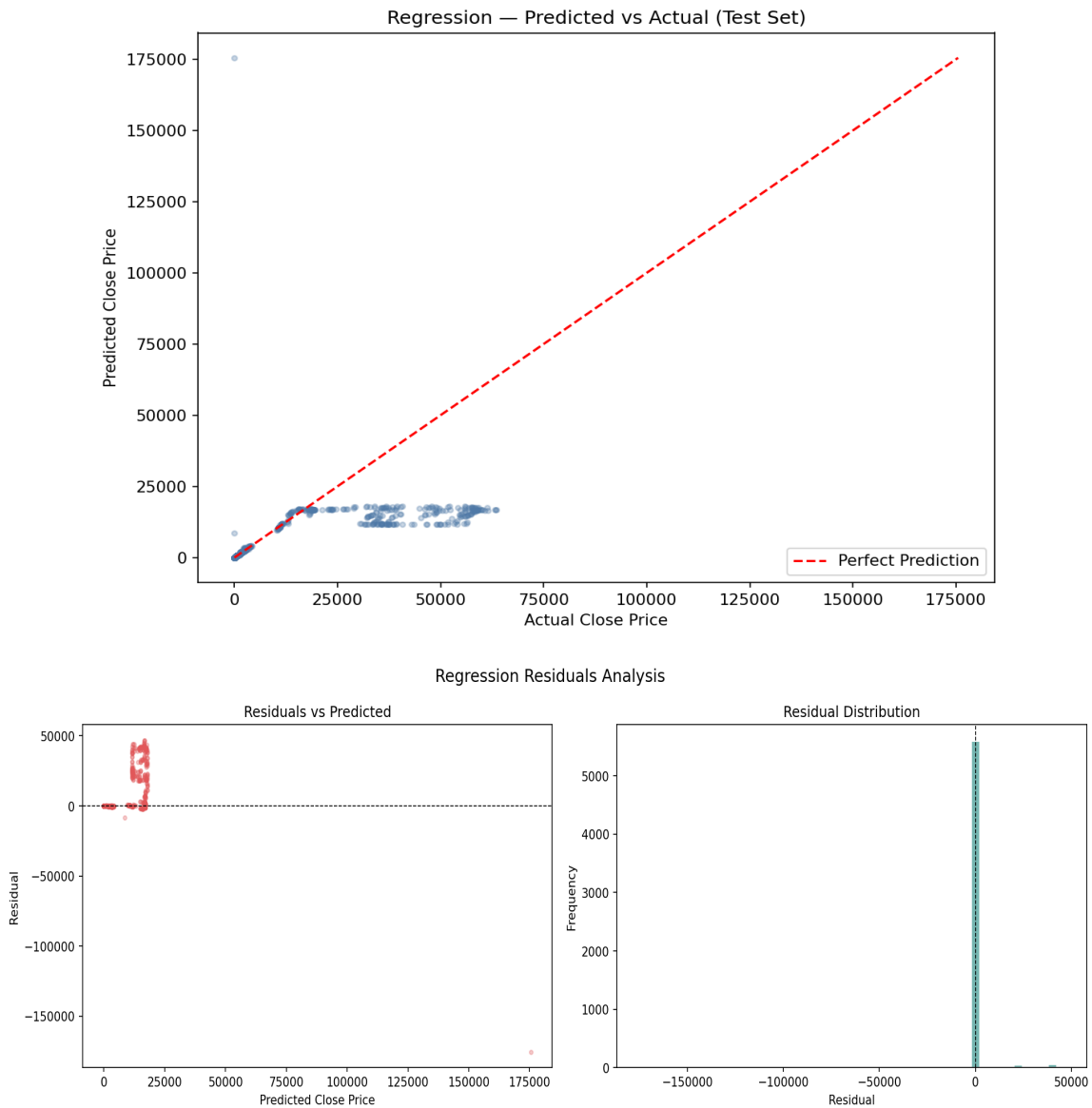


2. Regression Model

A RandomForestRegressor was trained to predict the next-day close price. Metrics include MAE, RMSE, MAPE, and R^2 .

Regression — Test Set Metrics

	MAE	RMSE	MAPE (%)	R^2	CV R^2 mean
RFRegressor	1061.8521	6201.1852	89019.7327	0.4625	-16951839.9842



3. Clustering Models

Four clustering configurations were evaluated using Silhouette Score, Davies-Bouldin Index, and Calinski-Harabasz Index. Features used: mean_return, volatility, sharpe_ratio, max_drawdown, avg_volume.

Clustering Quality Metrics

	K	Silhouette	Davies-Bouldin	Calinski-Harabasz
KMeans_K2	2.0	0.6726	0.2123	13.3969
KMeans_K4	4.0	0.5318	0.2291	19.0527
DBSCAN	1.0	nan	nan	nan
Agglomerative_K2	2.0	0.6726	0.2123	13.3969

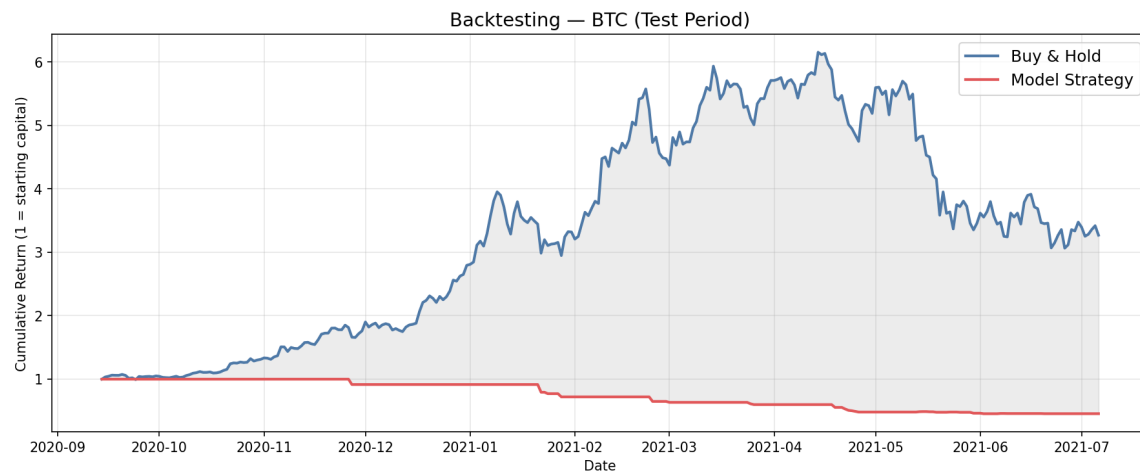


4. Backtesting Simulation

A simple simulation compares a strategy based on classifier predictions (buy when prediction=1, sell when prediction=0) against a passive buy & hold strategy over the test period.

Backtesting Results

	Test_Days	Days_In_Market	Days_Out_Market	BuyHold_Total_Return	Strategy_Total_Return	Strategy_vs_BuyHold_p
BTC	296.0	26.0	270.0	226.88	-54.41	-86.05
ETH	296.0	238.0	58.0	501.41	-17.13	-86.22



5. Consolidated Model Comparison

The table below summarises the key metric for each model type to enable a quick cross-comparison.

All Models at a Glance

	Type	Key Metric
LogisticRegression	Classification	ROC-AUC = 0.4769
RandomForestClassifier	Classification	ROC-AUC = 0.5244
XGBClassifier	Classification	ROC-AUC = 0.5305
RFRegressor	Regression	R ² = 0.4625
KMeans_K2	Clustering	Silhouette = 0.6726
KMeans_K4	Clustering	Silhouette = 0.5318
DBSCAN	Clustering	Silhouette = nan
Agglomerative_K2	Clustering	Silhouette = 0.6726

End of report. Generated automatically by src/scoring.py.